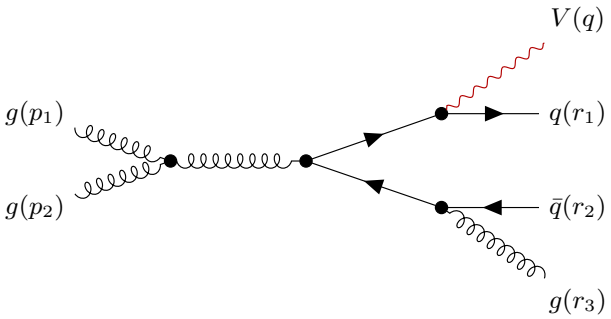
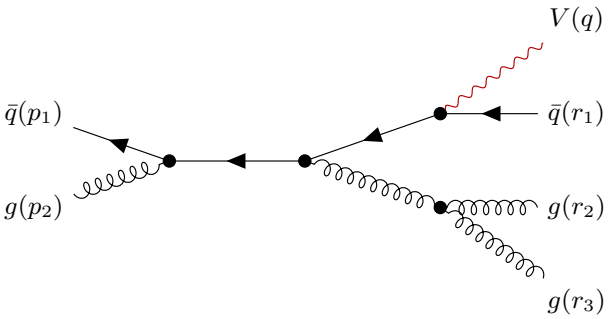
Triple-real channels: one representative diagram each

QGRAF diagrams (all legs incoming); initial state p_1, p_2 on the left, V and the final state r_1, r_2, r_3 on the right (final-state particles = conjugates of the QGRAF fields). Representative: first diagram with only 3-point vertices and V on the q line. Black: q, blue: Q, dotted: ghosts, curly: gluons, wavy: V. Arrows: fermion flow.

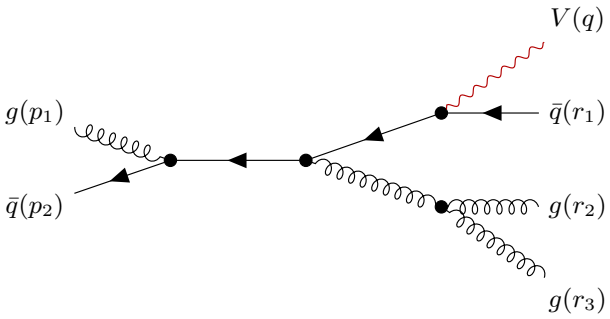
physical_qqbggg



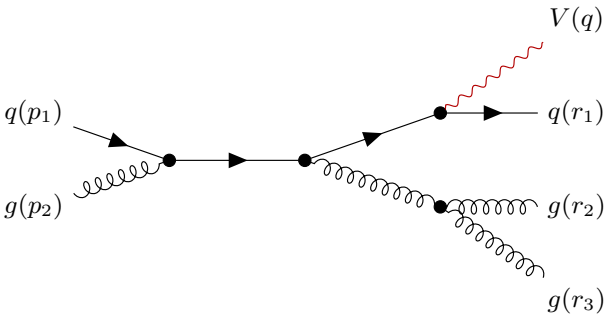
Vggqqbg: $gg \to V\, q\bar{q}g$
diagram 3 of 50, p1↔p2 symmetric



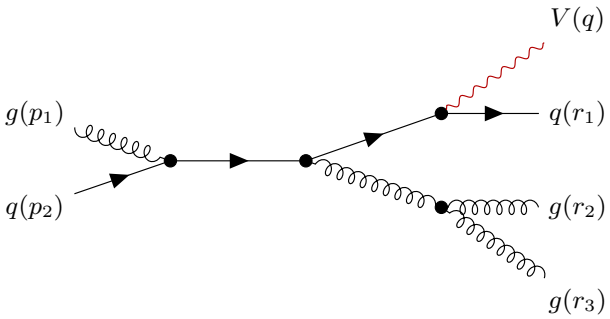
Vqbgqbgg: $\bar{q}g \to V\, \bar{q}gg$
diagram 3 of 50, p1↔p2 partner Vgqbqbgg



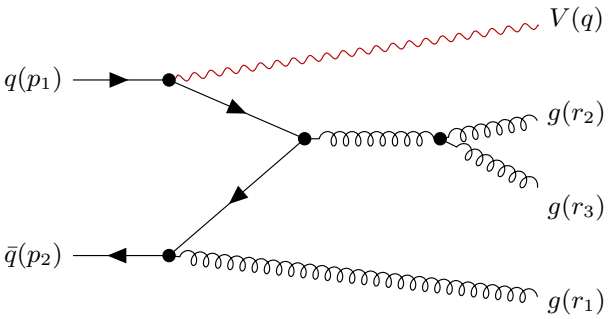
Vgqbqbgg: $g\bar{q} \to V\, \bar{q}gg$
diagram 3 of 50, p1↔p2 partner Vqbgqbgg



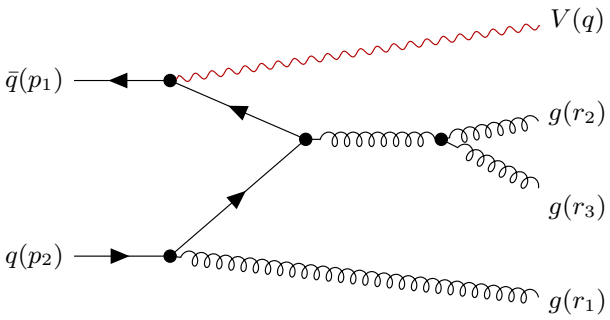
Vqgqgg: $qg \to V\, qgg$
diagram 3 of 50, p1↔p2 partner Vgqqgg



Vgqqgg: $gq \to V\, qgg$
diagram 3 of 50, p1↔p2 partner Vqgqgg

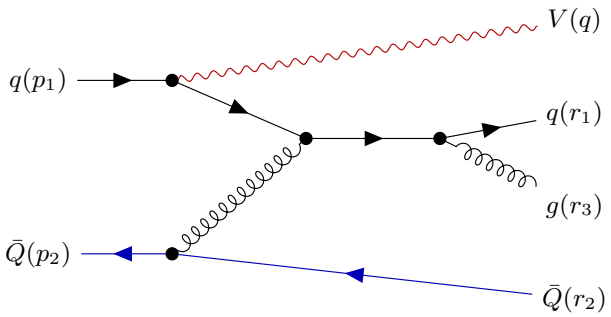


Vqqbggg: $q\bar{q} \to V\, ggg$
diagram 3 of 50, p1↔p2 partner Vqbqggg

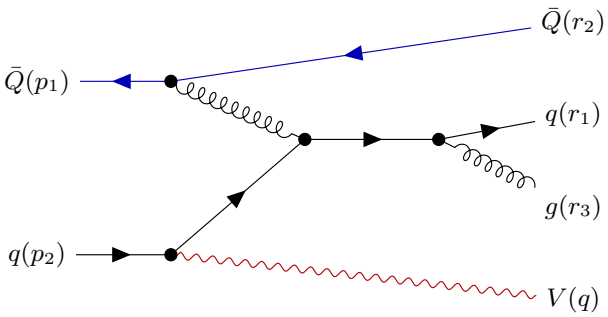


Vqbqggg: $\bar{q}q \to V\, ggg$
diagram 3 of 50, p1↔p2 partner Vqqbggg

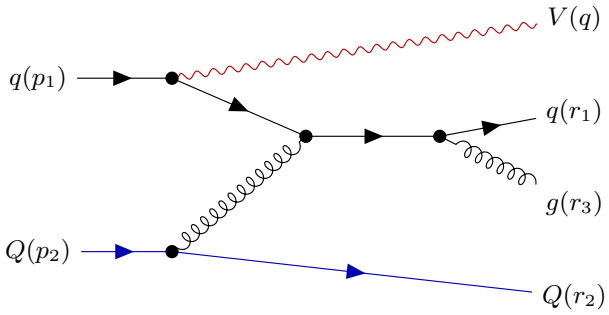
physical_qqbQ2Q2bg



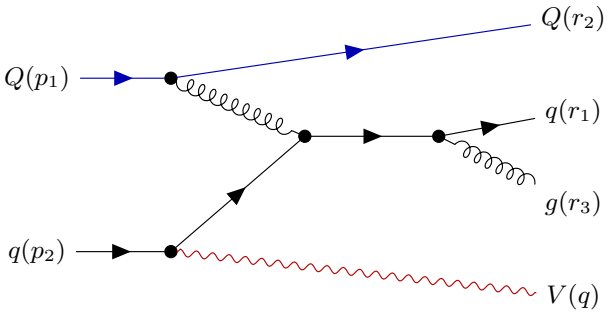
VqQ2bqQ2bg: $q\bar{Q} \to V\, q\bar{Q}g$
diagram 1 of 24, p1↔p2 partner VQ2bqqQ2bg



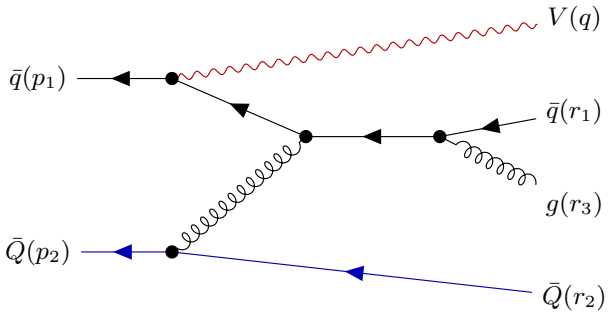
VQ2bqqQ2bg: $\bar{Q}q \to V\, q\bar{Q}g$
diagram 2 of 24, p1↔p2 partner VqQ2bqQ2bg



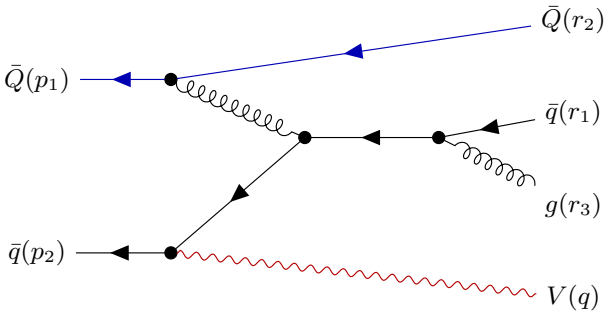
VqQ2qQ2g: $qQ \to V\, qQg$
diagram 1 of 24, p1↔p2 partner VQ2qqQ2g



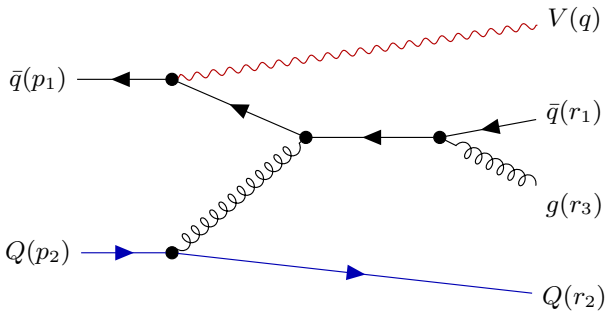
VQ2qqQ2g: $Qq \to V\, qQg$
diagram 2 of 24, p1↔p2 partner VqQ2qQ2g



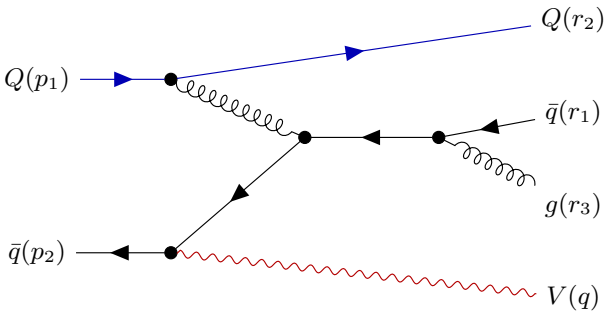
VqbQ2bqbQ2bg: $\bar{q}\bar{Q} \to V\, \bar{q}\bar{Q}g$
diagram 1 of 24, p1↔p2 partner VQ2bqbqbQ2bg



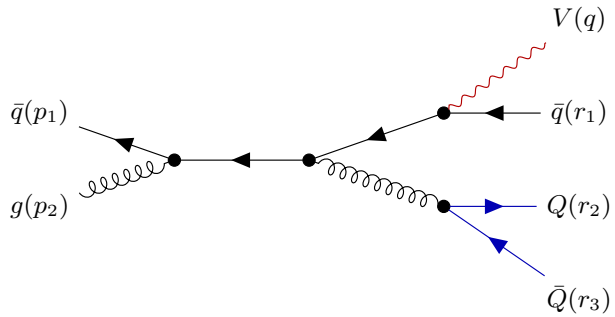
VQ2bqbqbQ2bg: $\bar{Q}\bar{q} \to V\, \bar{q}\bar{Q}g$
diagram 2 of 24, p1↔p2 partner VqbQ2bqbQ2bg



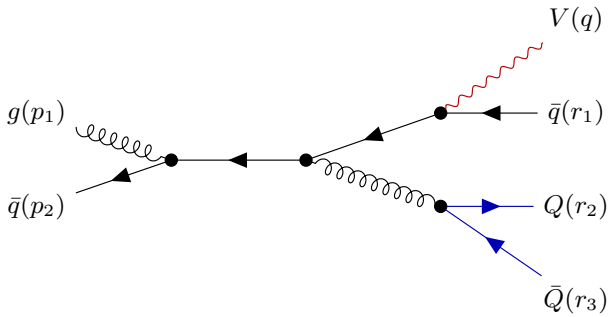
VqbQ2qbQ2g: $\bar{q}Q \to V\, \bar{q}Qg$
diagram 1 of 24, p1↔p2 partner VQ2qbqbQ2g



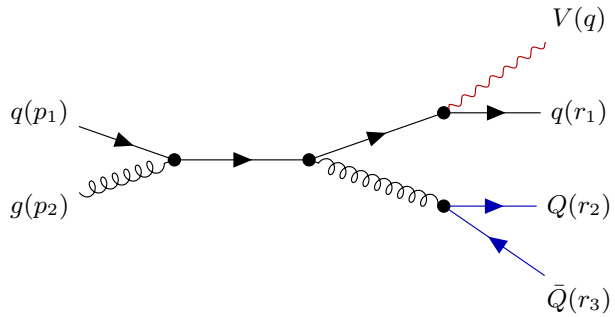
VQ2qbqbQ2g: $Q\bar{q} \to V\, \bar{q}Qg$
diagram 2 of 24, p1↔p2 partner VqbQ2qbQ2g



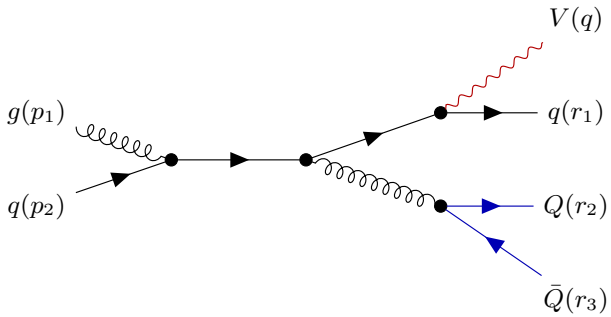
VqbgqbQ2Q2b: $\bar{q}g \to V\,\bar{q}Q\bar{Q}$
diagram 1 of 24, p1↔p2 partner VgqbqbQ2Q2b



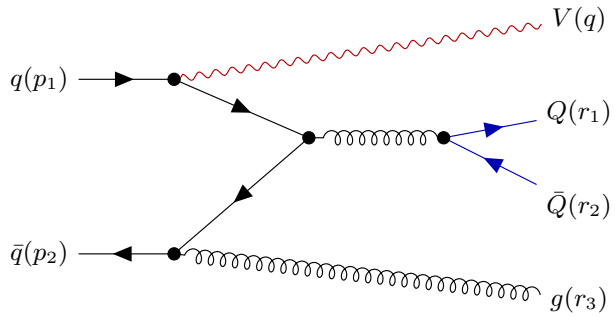
VgqbqbQ2Q2b: $g\bar{q} \to V\,\bar{q}Q\bar{Q}$
diagram 1 of 24, p1↔p2 partner VqbgqbQ2Q2b



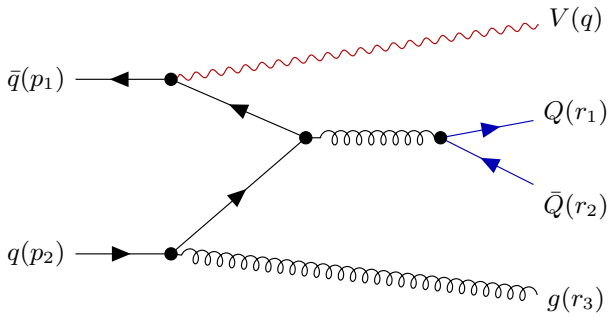
VqgqQ2Q2b: $qg \to V\,qQ\bar{Q}$
diagram 1 of 24, p1↔p2 partner VgqqQ2Q2b



VgqqQ2Q2b: $gq \to V\,qQ\bar{Q}$
diagram 1 of 24, p1↔p2 partner VqgqQ2Q2b

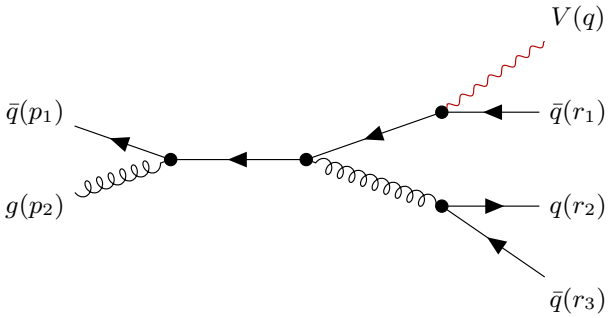


VqqbQ2Q2bg: $q\bar{q} \to V\,Q\bar{Q}g$
diagram 3 of 24, p1↔p2 partner VqbqQ2Q2bg

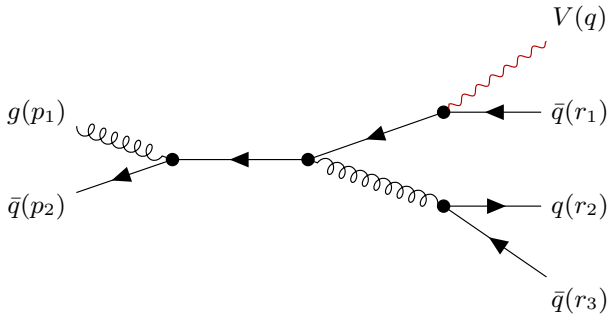


VqbqQ2Q2bg: $\bar{q}q \to V\,Q\bar{Q}g$
diagram 3 of 24, p1↔p2 partner VqqbQ2Q2bg

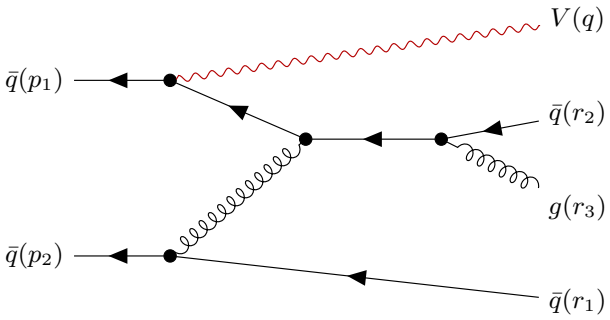
physical_qqbqqbg



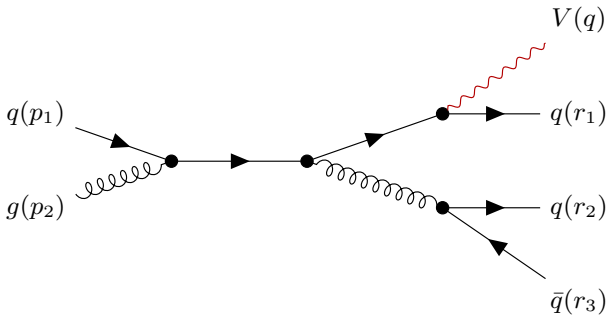
Vqbgqbqqb: $\bar{q}g \to V\,\bar{q}q\bar{q}$
diagram 1 of 48, p1↔p2 partner Vgqbqbqqb



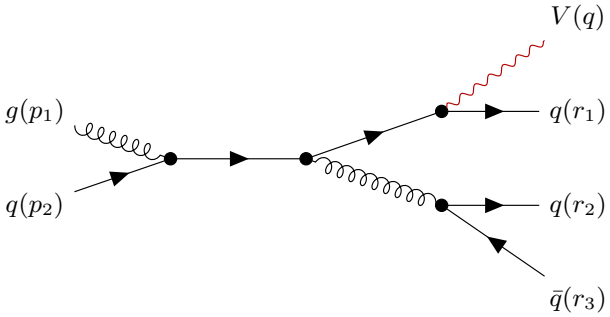
Vgqbqbqqb: $g\bar{q} \to V\,\bar{q}q\bar{q}$
diagram 1 of 48, p1↔p2 partner Vqbgqbqqb



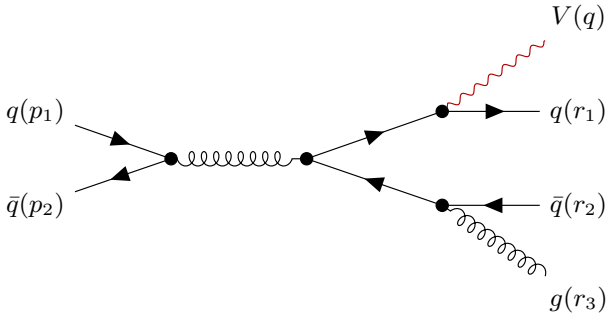
Vqbqbqbqbg: $\bar{q}\bar{q} \to V\,\bar{q}\bar{q}g$
diagram 1 of 48, p1↔p2 symmetric



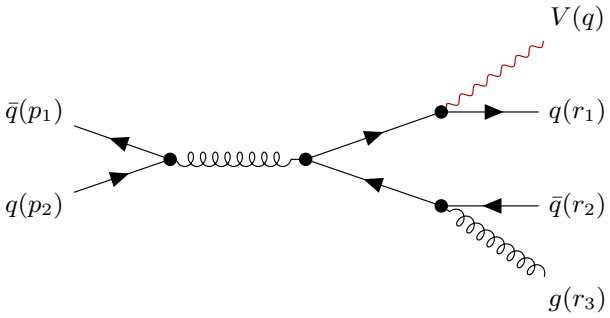
Vqgqqqb: $qg \to V\,qq\bar{q}$
diagram 1 of 48, p1↔p2 partner Vgqqqqb



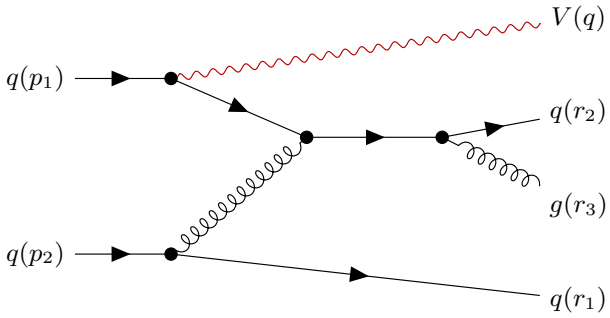
Vgqqqqb: $gq \to V\,qq\bar{q}$
diagram 1 of 48, p1↔p2 partner Vqgqqqb



Vqqbqqbg: $q\bar{q} \to V\,q\bar{q}g$
diagram 1 of 48, p1↔p2 partner Vqbqqqbg

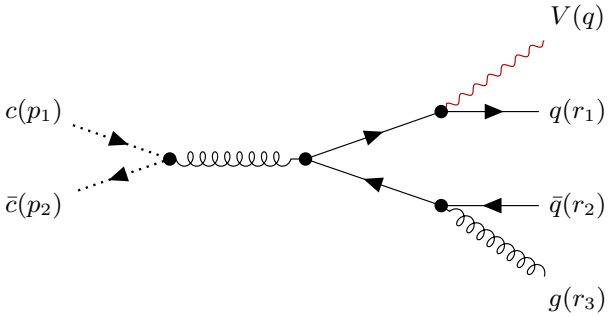


Vqbqqqbg: $\bar{q}q \to V\,q\bar{q}g$
diagram 1 of 48, p1↔p2 partner Vqqbqqbg

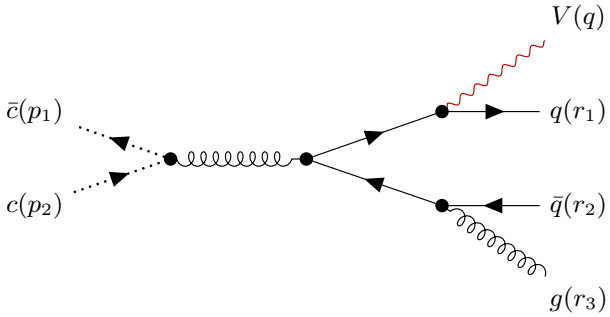


Vqqqqg: $qq \to V\,qqg$
diagram 1 of 48, p1↔p2 symmetric

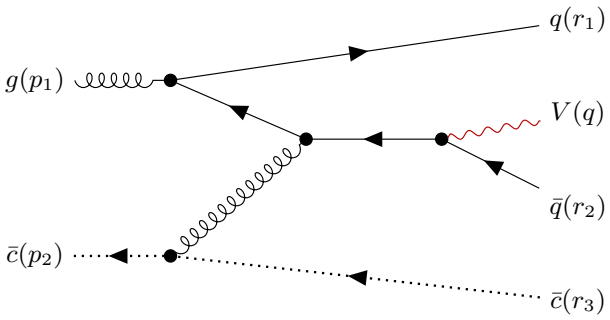
ghosts_qqbccbg



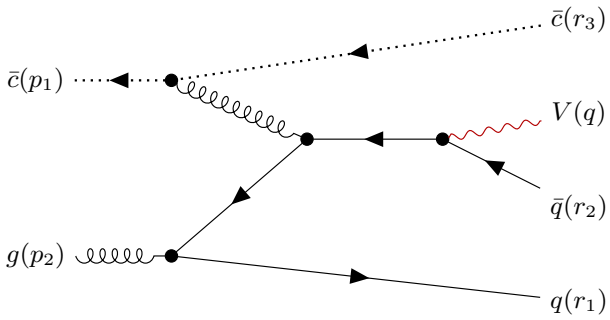
Vccbqqbg: $c\bar{c} \to V\,q\bar{q}g$
diagram 1 of 12, p1↔p2 partner Vcbcqqbg



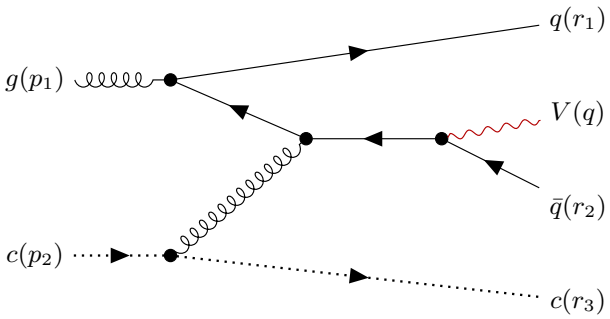
Vcbcqqbg: $\bar{c}c \to V\,q\bar{q}g$
diagram 1 of 12, p1↔p2 partner Vccbqqbg



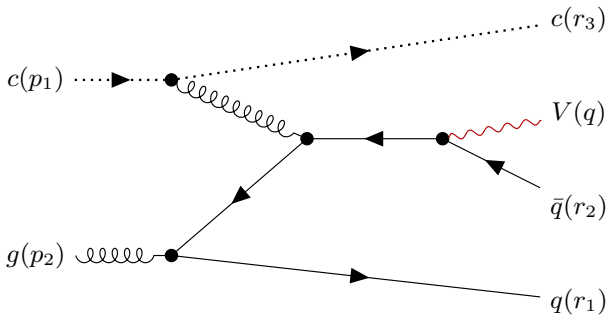
Vgcbqqbcb: $g\bar{c} \to V\, q\bar{q}\bar{c}$
diagram 1 of 12, p1↔p2 partner Vcbgqqbcb



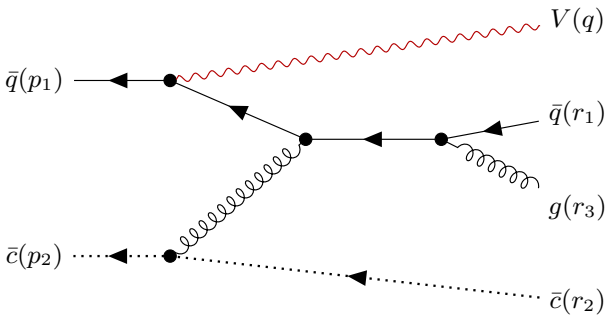
Vcbgqqbcb: $\bar{c}g \to V\, q\bar{q}\bar{c}$
diagram 1 of 12, p1↔p2 partner Vgcbqqbcb



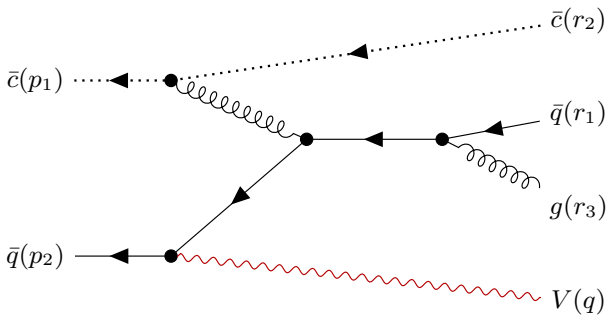
Vgcqqbc: $gc \to V\, q\bar{q}c$
diagram 1 of 12, p1↔p2 partner Vcgqqbc



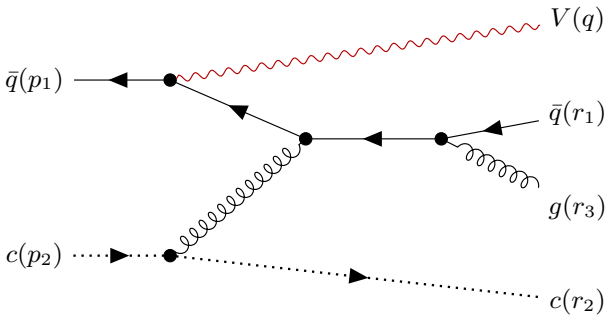
Vcgqqbc: $cg \to V\, q\bar{q}c$
diagram 1 of 12, p1↔p2 partner Vgcqqbc



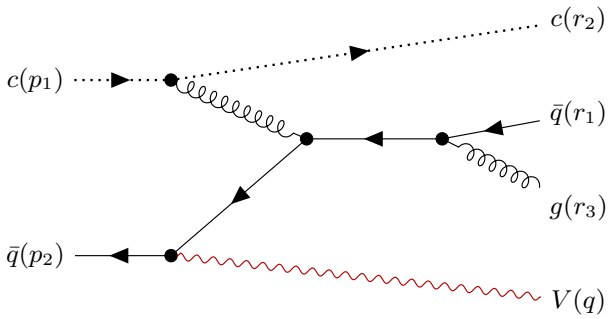
Vqbcbqbcbg: $\bar{q}\bar{c} \to V\, \bar{q}\bar{c}g$
diagram 1 of 12, p1↔p2 partner Vcbqbqbcbg



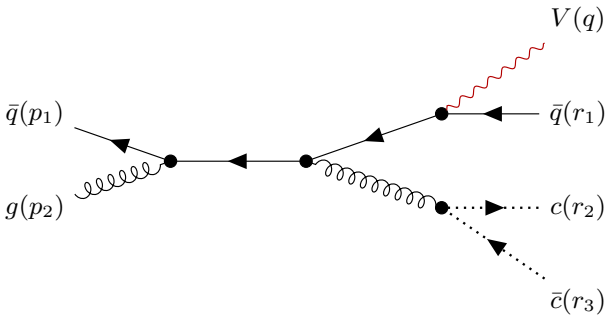
Vcbqbqbcbg: $\bar{c}\bar{q} \to V\, \bar{q}\bar{c}g$
diagram 1 of 12, p1↔p2 partner Vqbcbqbcbg



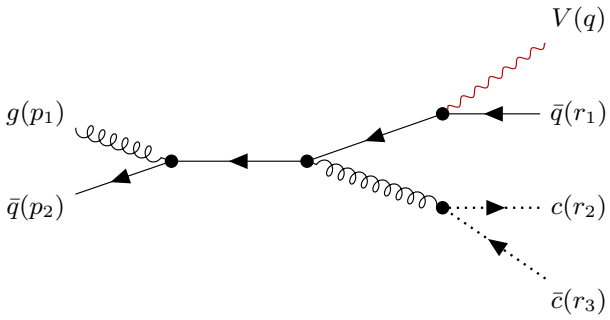
Vqbcqbcg: $\bar{q}c \to V\, \bar{q}cg$
diagram 1 of 12, p1↔p2 partner Vcqbqbcg



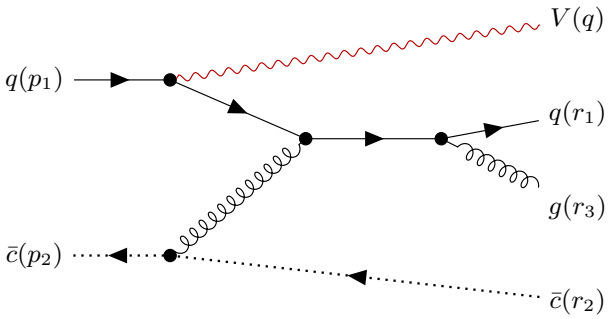
Vcqbqbcg: $c\bar{q} \to V\, \bar{q}cg$
diagram 1 of 12, p1↔p2 partner Vqbcqbcg



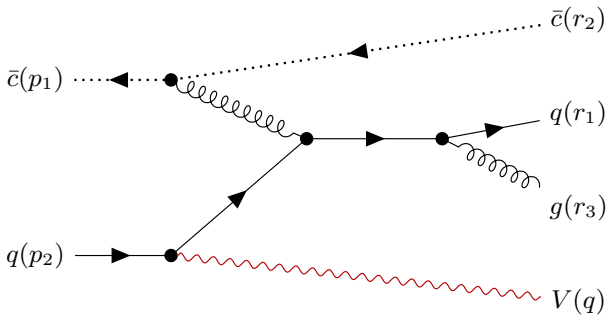
Vqbgqbccb: $\bar{q}g \to V\,\bar{q}c\bar{c}$
diagram 1 of 12, p1↔p2 partner Vgqbqbccb



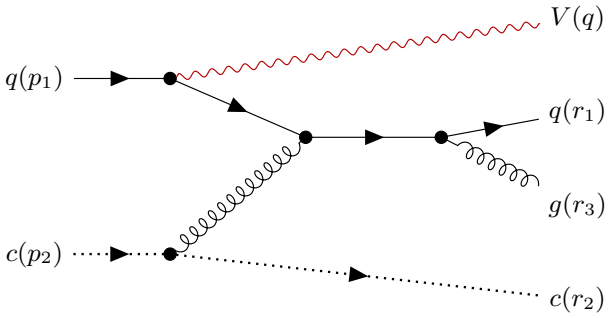
Vgqbqbccb: $g\bar{q} \to V\,\bar{q}c\bar{c}$
diagram 1 of 12, p1↔p2 partner Vqbgqbccb



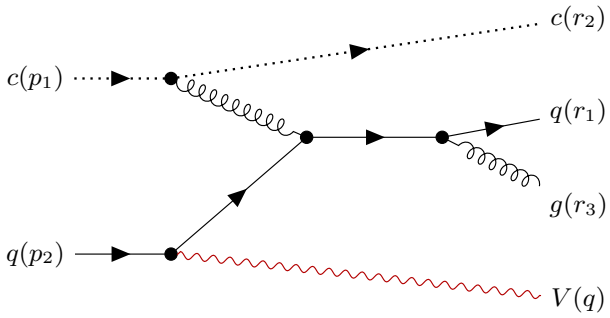
Vqcbqcbg: $q\bar{c} \to V\,q\bar{c}g$
diagram 1 of 12, p1↔p2 partner Vcbqqcbg



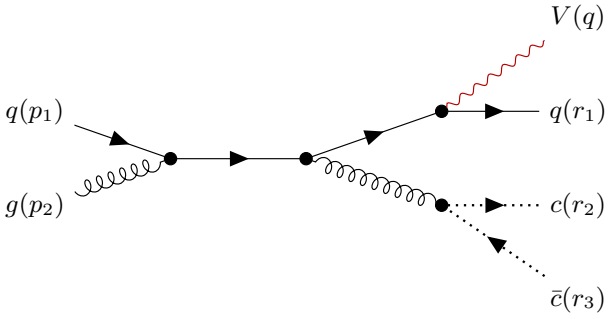
Vcbqqcbg: $\bar{c}q \to V\,q\bar{c}g$
diagram 1 of 12, p1↔p2 partner Vqcbqcbg



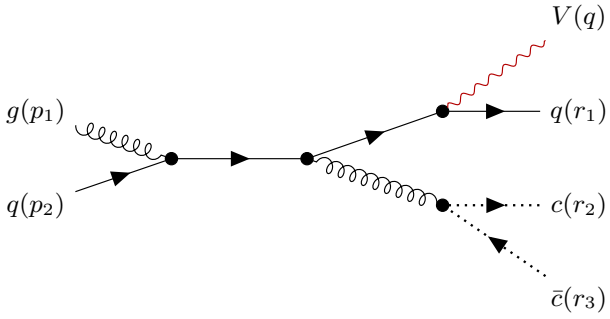
Vqcqcg: $qc \to V\,qcg$
diagram 1 of 12, p1↔p2 partner Vcqqcg



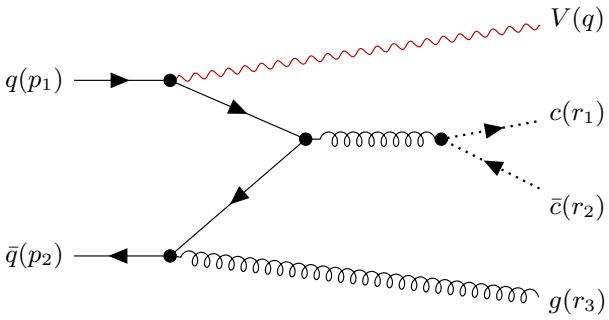
Vcqqcg: $cq \to V\,qcg$
diagram 1 of 12, p1↔p2 partner Vqcqcg



Vqgqccb: $qg \to V\,qc\bar{c}$
diagram 1 of 12, p1↔p2 partner Vgqqccb

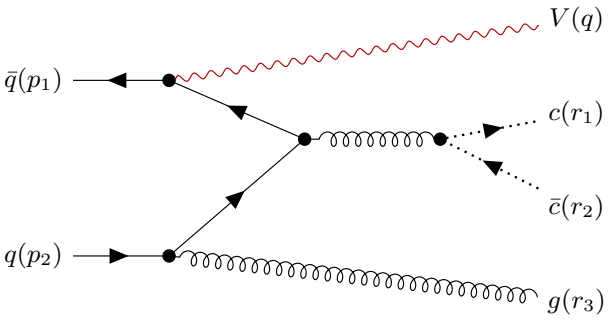


Vgqqccb: $gq \to V\,qc\bar{c}$
diagram 1 of 12, p1↔p2 partner Vqgqccb



Vqqbccbg: $q\bar{q} \to V\, c\bar{c}g$

diagram 1 of 12, p1↔p2 partner Vqbqccbg



Vqbqccbg: $\bar{q}q \to V\, c\bar{c}g$

diagram 1 of 12, p1↔p2 partner Vqqbccbg